UNIVERSITY OF RWANDA

COLLEGE OF SCIENCE & TECHNOLOGY

SCHOOL OF ICT

COMPUTER ENGENEERING

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MOBILE APPLICATION LAB 3 REPORT

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WanderNest

Travel App — Design & Architecture Report

Layout Choices · Widgets Catalog · Navigation Flow

1. Layout Choices

Home Screen

SliverAppBar	Collapsible hero header with gradient and decorative circles
CustomScrollView	Unified, performance-efficient infinite scroll
SliverGrid (2-col)	Responsive 0.72 aspect-ratio destination grid
Horizontal ListView	Featured carousel with fixed 220px width cards
AnimatedContainer	Pill-style bottom nav with expand animation

Detail Screen

Stack layout	Full-bleed image header with overlaid action buttons
SingleChildScrollView	Bouncing physics for immersive scroll experience
Rounded white card	Slides over image for content with 28px top radius
GridView (4-col)	Compact amenity badge grid, non-scrollable
Sticky bottom bar	Positioned price + Book Now; always accessible

Booking Screen

AppBar + back btn	Minimal header, centered title, card back button
SlideTransition	Slide-up + fade entrance animation on load
Section-card pattern	Each form section in its own rounded white card
Row date picker	Side-by-side check-in/check-out with night badge
Confirm Dialog	Full success state with booking summary and CTA

2. Core Design Principles

Depth & Elevation	Cards use multi-layer box shadows (blur 20, offset 8) to create clear visual hierarchy. Bottom bars float above content.
Color System	Primary indigo (#4A6CF7) with primaryLight tint for backgrounds. Accent coral (#FF6B6B) for notifications and favorites.
Typography	Poppins font family throughout — w800 for titles, w600 for subheads, w400 for body. Inter-line spacing 1.25–1.5.
Motion & Feedback	AnimatedContainer (200ms) for nav pills. ScaleTransition (elasticOut) on Book Now FAB. SlideTransition (easeOutCubic) on booking entry.

3. Widgets Catalog

DestinationCard `destination_card.dart`

Hero image card with gradient overlay, title, location chip, rating, and price. Used in both grid and carousel.

CategoryChip `category_chip.dart`

Pill-shaped filter chip with emoji icon. Selected state uses primary fill; unselected is white with border.

SectionHeader `section_header.dart`

Two-column row: bold section title (left) + tappable action label (right) with consistent 20px horizontal padding.

ImageSlider `image_slider.dart`

PageView-based full-bleed image carousel with dot indicator. Used for destination gallery (height 380px).

StarRatingBar `rating_bar.dart`

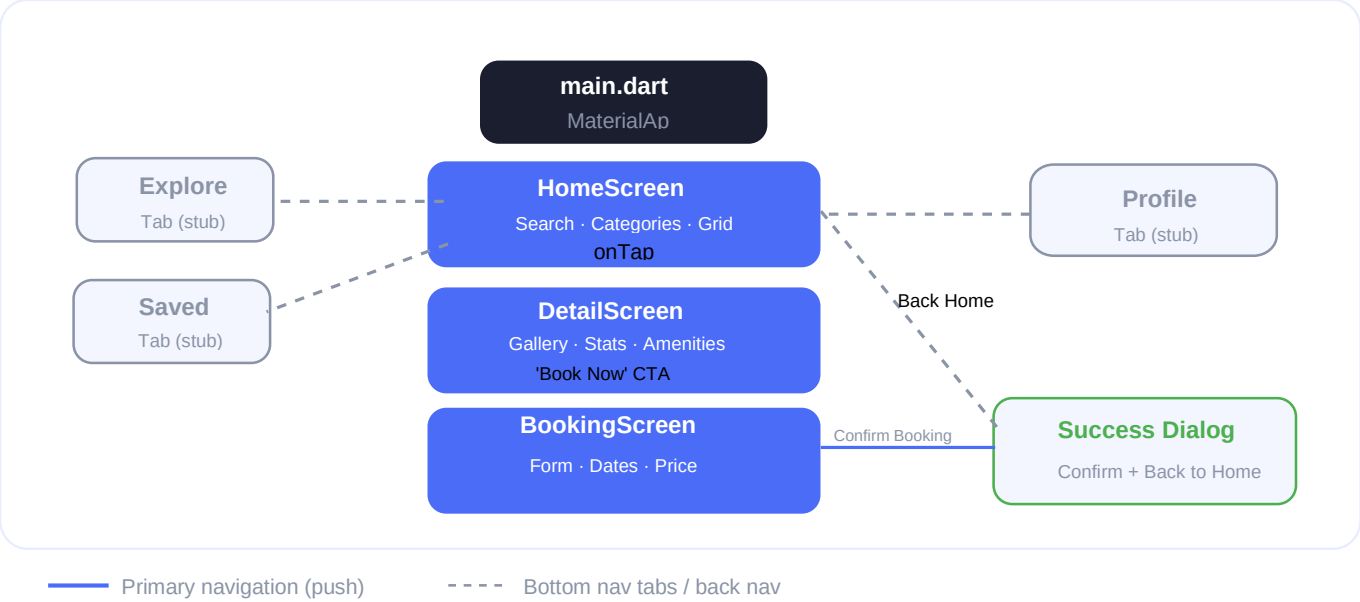
Renders 5 star icons (filled / half / empty) from a double rating value. Inline review count in grey.

AmenityBadge `amenity_badge.dart`

Compact icon + label tile for amenity grid. Derives icon from label string via mapping. 4-column layout.

4. Navigation Flow

The app follows a linear primary push-navigation pattern from Home through to booking confirmation, with stub tabs for secondary sections.

Screen	Route / Trigger	Destination
main.dart	App Entry Point	HomeScreen
HomeScreen	onTap destination	DetailScreen
DetailScreen	'Book Now' CTA	BookingScreen
BookingScreen	Confirm Booking	Success Dialog
Success Dialog	Back Home	HomeScreen
HomeScreen	Bottom Nav Tabs	Explore / Saved / Profile (stubs)

NAVIGATION FLOW DIAGRAM